

Supplementary Information for
“Depth Matters but the Final Layer Suffices: Leakage-Free
Evaluation of Layer Choice in Protein Language Models for
Mutation Effect Prediction”

S1. Assay composition of the benchmark

Table 1: Assay composition of the 63-dataset benchmark (SaProt LMDb snapshot, contained in ProteinGym v1.3). Multi-site variants (≥ 2 substitutions) dominate the row count; caps follow the fixed protocol (10,000 variants, then 5,000 rows, seed 42).

Assay	Variants total	Multi-site (%)	Rows after cap	Analyzed rows	Variant cap	Row ceiling
A0A1I9GEU1_NEIME_Kennouche_2019	922	0	922	922		
A0A2Z5U3Z0_9INFA_Doud_2016	10,715	0	10,715	5000	yes	yes
A4GRB6_PSEAI_Chen_2020	5,004	0	5,004	5000		yes
A4_HUMAN_Seuma_2021	14,483	97	28,498	5000	yes	yes
AACC1_PSEAI_Dandage_2018	1,801	0	1,801	1801		
ADRB2_HUMAN_Jones_2020	7,800	0	7,800	5000		yes
AMIE_PSEAE_Wrenbeck_2017	6,227	0	6,227	5000		yes
B3VI55_LIPST_Klesmith_2015	7,890	0	7,890	5000		yes
BLAT_ECOLX_Deng_2012	4,996	0	4,996	4996		
BLAT_ECOLX_Firnberg_2014	4,783	0	4,783	4783		
BLAT_ECOLX_Jacquier_2013	989	0	989	989		
BLAT_ECOLX_Stiffler_2015	4,996	0	4,996	4996		
CALM1_HUMAN>Weile_2017	1,813	0	1,813	1813		
CCDB_ECOLI_Adkar_2012	1,176	0	1,176	1176		
CCDB_ECOLI_Tripathi_2016	1,663	0	1,663	1663		
CP2C9_HUMAN_Amorosi_abundance_2021	6,370	0	6,370	5000		yes
CP2C9_HUMAN_Amorosi_activity_2021	6,142	0	6,142	5000		yes
DLG4_HUMAN_Faure_2021	6,976	82	12,672	5000		yes
DLG4_RAT_McLaughlin_2012	1,576	0	1,576	1576		
DYR_ECOLI_Thompson_plusLon_2019	2,363	0	2,363	2363		
ESTA_BACSU_Nutschel_2020	2,172	0	2,172	2172		
F7YBW8_MESOW_Aakre_2015	9,192	100	32,861	5000		yes
GAL4_YEAST_Kitzman_2015	1,195	0	1,195	1195		
GCN4_YEAST_Staller_induction_2018	2,638	99	45,025	5000		yes
GFP_AEQVI_Sarkisyan_2016	51,714	98	200,547	5000	yes	yes
GRB2_HUMAN_Faure_2021	63,366	98	125,698	5000	yes	yes
HIS7_YEAST_Pokusaeva_2019	496,137	100	3,455,486	5000	yes	yes
HSP82_YEAST_Flynn_2019	13,194	0	13,194	5000	yes	yes
HSP82_YEAST_Mishra_2016	4,323	0	4,323	4323		
IF1_ECOLI_Kelsic_2016	1,367	0	1,367	1367		
KKÄ2_KLEPN_Melnikov_2014	4,960	0	4,960	4960		
MK01_HUMAN_Brenan_2016	6,809	0	6,809	5000		yes
MSH2_HUMAN_Jia_2020	16,749	0	16,749	5000	yes	yes
MTH3_HAEAE_Rockah-Shmuel_2015	1,777	0	1,777	1777		
NUD15_HUMAN_Suiter_2020	2,844	0	2,844	2844		
P53_HUMAN_Giacomelli_NULL_Etoposide_2018	7,467	0	7,467	5000		yes
P53_HUMAN_Giacomelli_NULL_Nutlin_2018	7,467	0	7,467	5000		yes
P53_HUMAN_Giacomelli_WT_Nutlin_2018	7,467	0	7,467	5000		yes
P53_HUMAN_Kotler_2018	1,048	0	1,048	1048		
P84I26_THETH_Chan_2017	1,519	0	1,519	1519		
PABP_YEAST_Melamed_2013	37,708	97	74,229	5000	yes	yes
PTEN_HUMAN_Matreyek_2021	5,083	0	5,083	5000		yes
PTEN_HUMAN_Mighell_2018	7,260	0	7,260	5000		yes
Q59976_STRSQ_Romero_2015	2,999	0	2,999	2999		
RASH_HUMAN_Bandaru_2017	3,134	0	3,134	3134		
RL401_YEAST_Mavor_2016	1,253	0	1,253	1253		
RL401_YEAST_Roscoe_2013	1,195	0	1,195	1195		
RL401_YEAST_Roscoe_2014	1,380	0	1,380	1380		
SC6A4_HUMAN_Young_2021	11,576	0	11,576	5000	yes	yes
SPG1_STRSG_Olson_2014	536,962	100	1,072,879	5000	yes	yes
SRC_HUMAN_Ahler_CD_2019	3,372	0	3,372	3372		
SUMO1_HUMAN>Weile_2017	1,700	0	1,700	1700		
SYUA_HUMAN_Newberry_2020	2,497	0	2,497	2497		
TADBP_HUMAN_Bolognesi_2019	1,196	0	1,196	1196		
TPK1_HUMAN>Weile_2017	3,181	0	3,181	3181		
TPMT_HUMAN_Matreyek_2018	3,648	0	3,648	3648		
TPOR_HUMAN_Bridgford_S505N_2020	562	0	562	562		
TRPC_SACS2_Chan_2017	1,519	0	1,519	1519		
TRPC_THEMA_Chan_2017	1,519	0	1,519	1519		
UBC9_HUMAN>Weile_2017	2,563	0	2,563	2563		
VKOR1_HUMAN_Chiasson_abundance_2020	2,695	0	2,695	2695		
VKOR1_HUMAN_Chiasson_activity_2020	697	0	697	697		
YAP1_HUMAN_Araya_2012	10,075	96	19,788	5000	yes	yes
Total / pooled	1,445,864	84.6	5,284,296	213,393	11	26

S2. Full 33-layer mean Spearman ρ (combined features)

Table 2: ESM-2, masked_wt arm: per-layer mean ρ by split (best layer per column in bold). $n = 63$ (contiguous 57).

Layer	Random	Modulo	Contiguous	Pos.-grouped
0	0.601	0.232	nan	0.224
1	0.601	0.235	nan	0.219
2	0.602	0.262	nan	0.252
3	0.602	0.275	nan	0.267
4	0.603	0.285	nan	0.276
5	0.602	0.301	nan	0.290
6	0.603	0.307	nan	0.298
7	0.603	0.312	nan	0.303
8	0.603	0.316	nan	0.306
9	0.603	0.322	nan	0.308
10	0.603	0.324	nan	0.315
11	0.603	0.334	nan	0.323
12	0.603	0.332	nan	0.326
13	0.603	0.339	nan	0.327
14	0.603	0.336	nan	0.328
15	0.603	0.340	nan	0.331
16	0.603	0.344	nan	0.333
17	0.604	0.348	nan	0.341
18	0.604	0.347	nan	0.339
19	0.604	0.350	nan	0.341
20	0.604	0.359	nan	0.352
21	0.604	0.364	nan	0.354
22	0.604	0.364	nan	0.355
23	0.604	0.368	nan	0.361
24	0.604	0.377	nan	0.369
25	0.604	0.387	nan	0.378
26	0.604	0.398	nan	0.389
27	0.605	0.406	nan	0.399
28	0.605	0.424	nan	0.414
29	0.605	0.435	nan	0.426
30	0.606	0.449	nan	0.440
31	0.606	0.453	nan	0.446
32	0.606	0.459	nan	0.450

Table 3: ESM-2, unmasked_wt arm: per-layer mean ρ by split (best layer per column in bold). $n = 63$ (contiguous 57).

Layer	Random	Modulo	Contiguous	Pos.-grouped
0	0.601	0.191	nan	0.188
1	0.602	0.257	nan	0.244
2	0.602	0.289	nan	0.281
3	0.603	0.306	nan	0.299
4	0.603	0.316	nan	0.305
5	0.603	0.331	nan	0.319
6	0.603	0.334	nan	0.323
7	0.603	0.336	nan	0.325
8	0.603	0.342	nan	0.329
9	0.603	0.343	nan	0.328
10	0.604	0.346	nan	0.335
11	0.604	0.349	nan	0.337
12	0.604	0.349	nan	0.338
13	0.603	0.350	nan	0.337
14	0.603	0.345	nan	0.338
15	0.603	0.351	nan	0.342
16	0.604	0.356	nan	0.347
17	0.604	0.361	nan	0.354
18	0.604	0.359	nan	0.351
19	0.604	0.361	nan	0.354
20	0.604	0.368	nan	0.363
21	0.604	0.374	nan	0.365
22	0.604	0.374	nan	0.367
23	0.604	0.381	nan	0.372
24	0.604	0.390	nan	0.380
25	0.604	0.398	nan	0.389
26	0.605	0.412	nan	0.403
27	0.605	0.422	nan	0.414
28	0.605	0.437	nan	0.428
29	0.606	0.448	nan	0.440
30	0.606	0.463	nan	0.453
31	0.606	0.465	nan	0.457
32	0.606	0.470	nan	0.461

Table 4: ESM-2, mutant arm: per-layer mean ρ by split (best layer per column in bold). $n = 63$ (contiguous 57).

Layer	Random	Modulo	Contiguous	Pos.-grouped
0	0.510	0.390	nan	0.381
1	0.615	0.311	nan	0.290
2	0.630	0.317	nan	0.309
3	0.639	0.324	nan	0.320
4	0.645	0.327	nan	0.318
5	0.647	0.340	nan	0.331
6	0.650	0.350	nan	0.339
7	0.649	0.362	nan	0.353
8	0.650	0.367	nan	0.352
9	0.649	0.369	nan	0.354
10	0.652	0.381	nan	0.370
11	0.652	0.391	nan	0.379
12	0.656	0.384	nan	0.377
13	0.659	0.372	nan	0.366
14	0.658	0.359	nan	0.351
15	0.658	0.358	nan	0.353
16	0.660	0.363	nan	0.356
17	0.660	0.371	nan	0.360
18	0.660	0.366	nan	0.359
19	0.660	0.367	nan	0.357
20	0.660	0.368	nan	0.359
21	0.660	0.363	nan	0.352
22	0.661	0.359	nan	0.354
23	0.662	0.363	nan	0.357
24	0.662	0.370	nan	0.359
25	0.664	0.370	nan	0.360
26	0.666	0.384	nan	0.378
27	0.670	0.402	nan	0.397
28	0.673	0.423	nan	0.413
29	0.676	0.451	nan	0.445
30	0.679	0.482	nan	0.476
31	0.679	0.483	nan	0.477
32	0.681	0.479	nan	0.475

Table 5: SaProt (mask), masked_wt arm: per-layer mean ρ by split (best layer per column in bold). $n = 63$ (contiguous 57).

Layer	Random	Modulo	Contiguous	Pos.-grouped
0	0.600	0.241	nan	0.229
1	0.602	0.230	nan	0.221
2	0.602	0.250	nan	0.246
3	0.602	0.262	nan	0.255
4	0.602	0.285	nan	0.275
5	0.602	0.292	nan	0.285
6	0.602	0.307	nan	0.300
7	0.603	0.317	nan	0.311
8	0.603	0.335	nan	0.327
9	0.603	0.339	nan	0.332
10	0.603	0.342	nan	0.335
11	0.603	0.352	nan	0.344
12	0.602	0.358	nan	0.350
13	0.602	0.359	nan	0.353
14	0.602	0.364	nan	0.360
15	0.602	0.366	nan	0.362
16	0.602	0.370	nan	0.364
17	0.602	0.371	nan	0.367
18	0.603	0.374	nan	0.368
19	0.602	0.374	nan	0.368
20	0.602	0.374	nan	0.370
21	0.602	0.374	nan	0.370
22	0.602	0.377	nan	0.373
23	0.602	0.381	nan	0.377
24	0.602	0.379	nan	0.376
25	0.602	0.378	nan	0.374
26	0.602	0.383	nan	0.379
27	0.603	0.387	nan	0.384
28	0.603	0.389	nan	0.383
29	0.603	0.388	nan	0.383
30	0.603	0.391	nan	0.383
31	0.603	0.391	nan	0.385
32	0.603	0.386	nan	0.382

Table 6: SaProt (mask), unmasked_wt arm: per-layer mean ρ by split (best layer per column in bold). $n = 63$ (contiguous 57).

Layer	Random	Modulo	Contiguous	Pos.-grouped
0	0.601	0.234	nan	0.227
1	0.602	0.262	nan	0.250
2	0.602	0.292	nan	0.286
3	0.602	0.307	nan	0.303
4	0.602	0.328	nan	0.319
5	0.603	0.329	nan	0.322
6	0.603	0.339	nan	0.332
7	0.603	0.349	nan	0.343
8	0.603	0.362	nan	0.356
9	0.603	0.368	nan	0.362
10	0.603	0.369	nan	0.363
11	0.603	0.374	nan	0.368
12	0.603	0.383	nan	0.375
13	0.603	0.386	nan	0.379
14	0.603	0.391	nan	0.383
15	0.603	0.396	nan	0.387
16	0.603	0.397	nan	0.386
17	0.603	0.400	nan	0.389
18	0.603	0.401	nan	0.389
19	0.603	0.401	nan	0.389
20	0.603	0.403	nan	0.392
21	0.603	0.405	nan	0.395
22	0.603	0.408	nan	0.398
23	0.603	0.409	nan	0.400
24	0.603	0.406	nan	0.399
25	0.603	0.408	nan	0.400
26	0.603	0.410	nan	0.402
27	0.604	0.411	nan	0.403
28	0.604	0.411	nan	0.401
29	0.604	0.412	nan	0.401
30	0.604	0.412	nan	0.401
31	0.604	0.411	nan	0.403
32	0.604	0.406	nan	0.400

Table 7: SaProt (mask), mutant arm: per-layer mean ρ by split (best layer per column in bold). $n = 63$ (contiguous 57).

Layer	Random	Modulo	Contiguous	Pos.-grouped
0	0.537	0.377	nan	0.361
1	0.611	0.313	nan	0.303
2	0.617	0.330	nan	0.323
3	0.620	0.336	nan	0.328
4	0.628	0.350	nan	0.340
5	0.630	0.354	nan	0.348
6	0.636	0.367	nan	0.359
7	0.637	0.375	nan	0.367
8	0.642	0.394	nan	0.387
9	0.646	0.399	nan	0.394
10	0.647	0.397	nan	0.392
11	0.648	0.407	nan	0.399
12	0.647	0.412	nan	0.404
13	0.648	0.415	nan	0.405
14	0.649	0.419	nan	0.412
15	0.649	0.423	nan	0.415
16	0.650	0.428	nan	0.417
17	0.650	0.431	nan	0.422
18	0.649	0.435	nan	0.425
19	0.649	0.434	nan	0.428
20	0.649	0.433	nan	0.428
21	0.650	0.438	nan	0.431
22	0.651	0.446	nan	0.439
23	0.652	0.452	nan	0.446
24	0.650	0.451	nan	0.449
25	0.650	0.448	nan	0.441
26	0.651	0.454	nan	0.448
27	0.650	0.458	nan	0.454
28	0.650	0.464	nan	0.458
29	0.650	0.461	nan	0.457
30	0.650	0.465	nan	0.461
31	0.650	0.462	nan	0.460
32	0.651	0.460	nan	0.458

Table 8: SaProt (3Di), masked_wt arm: per-layer mean ρ by split (best layer per column in bold). $n = 63$ (contiguous 57).

Layer	Random	Modulo	Contiguous	Pos.-grouped
0	0.601	0.274	nan	0.251
1	0.603	0.299	nan	0.289
2	0.603	0.337	nan	0.326
3	0.603	0.346	nan	0.336
4	0.603	0.364	nan	0.357
5	0.604	0.372	nan	0.365
6	0.604	0.379	nan	0.372
7	0.604	0.387	nan	0.378
8	0.604	0.398	nan	0.390
9	0.604	0.398	nan	0.392
10	0.604	0.399	nan	0.393
11	0.604	0.407	nan	0.399
12	0.605	0.411	nan	0.404
13	0.604	0.416	nan	0.408
14	0.605	0.420	nan	0.413
15	0.605	0.421	nan	0.414
16	0.605	0.424	nan	0.417
17	0.605	0.426	nan	0.420
18	0.605	0.427	nan	0.421
19	0.605	0.428	nan	0.422
20	0.605	0.431	nan	0.425
21	0.605	0.433	nan	0.429
22	0.605	0.436	nan	0.432
23	0.605	0.439	nan	0.434
24	0.605	0.442	nan	0.437
25	0.605	0.443	nan	0.438
26	0.605	0.447	nan	0.441
27	0.605	0.452	nan	0.445
28	0.605	0.455	nan	0.448
29	0.605	0.456	nan	0.450
30	0.605	0.459	nan	0.454
31	0.606	0.461	nan	0.454
32	0.606	0.460	nan	0.453

Table 9: SaProt (3Di), unmasked_wt arm: per-layer mean ρ by split (best layer per column in bold). $n = 63$ (contiguous 57).

Layer	Random	Modulo	Contiguous	Pos.-grouped
0	0.602	0.276	nan	0.263
1	0.603	0.333	nan	0.314
2	0.604	0.362	nan	0.346
3	0.604	0.376	nan	0.361
4	0.604	0.384	nan	0.373
5	0.604	0.393	nan	0.383
6	0.604	0.396	nan	0.385
7	0.604	0.402	nan	0.393
8	0.604	0.410	nan	0.401
9	0.604	0.413	nan	0.404
10	0.605	0.415	nan	0.406
11	0.605	0.422	nan	0.412
12	0.605	0.427	nan	0.417
13	0.605	0.431	nan	0.422
14	0.605	0.436	nan	0.428
15	0.605	0.439	nan	0.430
16	0.605	0.443	nan	0.435
17	0.605	0.444	nan	0.436
18	0.605	0.447	nan	0.439
19	0.605	0.448	nan	0.440
20	0.605	0.451	nan	0.443
21	0.605	0.454	nan	0.446
22	0.606	0.456	nan	0.449
23	0.606	0.458	nan	0.451
24	0.606	0.460	nan	0.453
25	0.606	0.462	nan	0.454
26	0.606	0.464	nan	0.457
27	0.606	0.468	nan	0.460
28	0.606	0.469	nan	0.461
29	0.605	0.469	nan	0.463
30	0.605	0.471	nan	0.467
31	0.605	0.473	nan	0.467
32	0.605	0.466	nan	0.459

Table 10: SaProt (3Di), mutant arm: per-layer mean ρ by split (best layer per column in bold). $n = 63$ (contiguous 57).

Layer	Random	Modulo	Contiguous	Pos.-grouped
0	0.578	0.365	nan	0.357
1	0.629	0.378	nan	0.364
2	0.640	0.402	nan	0.393
3	0.649	0.411	nan	0.401
4	0.653	0.415	nan	0.409
5	0.656	0.422	nan	0.417
6	0.660	0.431	nan	0.425
7	0.662	0.429	nan	0.424
8	0.665	0.436	nan	0.429
9	0.667	0.437	nan	0.432
10	0.669	0.432	nan	0.426
11	0.671	0.443	nan	0.436
12	0.671	0.447	nan	0.438
13	0.671	0.448	nan	0.437
14	0.671	0.454	nan	0.445
15	0.672	0.462	nan	0.455
16	0.672	0.467	nan	0.461
17	0.674	0.471	nan	0.465
18	0.674	0.468	nan	0.467
19	0.675	0.470	nan	0.465
20	0.677	0.473	nan	0.470
21	0.678	0.474	nan	0.475
22	0.679	0.491	nan	0.484
23	0.681	0.494	nan	0.487
24	0.682	0.496	nan	0.492
25	0.683	0.501	nan	0.496
26	0.684	0.505	nan	0.506
27	0.684	0.508	nan	0.508
28	0.684	0.518	nan	0.520
29	0.684	0.520	nan	0.521
30	0.682	0.530	nan	0.532
31	0.684	0.531	nan	0.532
32	0.683	0.525	nan	0.524

S3. Per-dataset results (mutant arm, modulo split)

Table 11: Per-dataset mutant-arm probing (modulo split, combined features): best layer (and ρ) and final-layer (L32) ρ .

Assay	n rows	ESM-2			SaProt (3Di)		
		best L	best ρ	L32 ρ	best L	best ρ	L32 ρ
A0A1I9GEU1_NEIME_Kennouche_2019	922	0	0.087	-0.077	3	0.004	-0.006
A0A2Z5U3Z0_9INFA_Doud_2016	5000	0	0.442	0.415	23	0.457	0.361
A4GRB6_PSEAI_Chen_2020	5000	32	0.548	0.548	31	0.751	0.677
A4_HUMAN_Seuma_2021	5000	15	0.674	0.623	3	0.677	0.656
AACC1_PSEAI_Dandage_2018	1801	29	0.470	0.426	32	0.549	0.549
ADRB2_HUMAN_Jones_2020	5000	31	0.507	0.506	31	0.527	0.526
AMIE_PSEAE_Wrenbeck_2017	5000	31	0.574	0.519	31	0.640	0.631
B3VI55_LIPST_Klesmith_2015	5000	31	0.562	0.477	32	0.592	0.592
BLAT_ECOLX_Deng_2012	4996	29	0.417	0.338	29	0.551	0.544
BLAT_ECOLX_Firnberg_2014	4783	31	0.656	0.600	30	0.799	0.768
BLAT_ECOLX_Jacquier_2013	989	31	0.581	0.559	29	0.739	0.729
BLAT_ECOLX_Stiffler_2015	4996	31	0.668	0.630	31	0.783	0.720
CALM1_HUMAN>Weile_2017	1813	30	0.154	0.132	6	0.141	0.126
CCDB_ECOLI_Adkar_2012	1176	30	0.613	0.580	30	0.694	0.639
CCDB_ECOLI_Tripathi_2016	1663	30	0.459	0.443	30	0.546	0.442
CP2C9_HUMAN_Amorosi_abundance_2021	5000	32	0.555	0.555	32	0.671	0.671
CP2C9_HUMAN_Amorosi_activity_2021	5000	31	0.598	0.583	29	0.631	0.601
DLG4_HUMAN_Faure_2021	5000	6	0.729	0.654	23	0.670	0.657
DLG4_RAT_McLaughlin_2012	1576	29	0.471	0.404	14	0.530	0.503
DYR_ECOLI_Thompson_plusLon_2019	2363	30	0.371	0.342	25	0.455	0.407
ESTA_BACSU_Nutschel_2020	2172	17	0.349	0.320	32	0.478	0.478
F7YBW8_MESOW_Aakre_2015	5000	12	0.299	0.238	19	0.255	0.231
GAL4_YEAST_Kitzman_2015	1195	30	0.471	0.451	32	0.448	0.448
GCN4_YEAST_Staller_induction_2018	5000	8	0.692	0.649	2	0.659	0.614
GFP_AEQVI_Sarkisyan_2016	5000	13	0.494	0.467	32	0.488	0.488
GRB2_HUMAN_Faure_2021	5000	8	0.656	0.591	32	0.629	0.629
HIS7_YEAST_Pokusaeva_2019	5000	12	0.629	0.605	32	0.649	0.649
HSP82_YEAST_Flynn_2019	5000	29	0.453	0.449	23	0.476	0.439
HSP82_YEAST_Mishra_2016	4323	27	0.507	0.465	31	0.501	0.486
IF1_ECOLI_Kelsic_2016	1367	31	0.334	0.316	17	0.485	0.245
KKA2_KLEPN_Melnikov_2014	4960	32	0.640	0.640	23	0.654	0.651
MK01_HUMAN_Brenan_2016	5000	0	0.517	0.360	10	0.353	0.342
MSH2_HUMAN_Jia_2020	5000	31	0.521	0.519	22	0.468	0.446
MTH3_HAEAE_Rockah-Shmuel_2015	1777	31	0.627	0.618	31	0.667	0.665
NUD15_HUMAN_Suiter_2020	2844	27	0.448	0.386	31	0.674	0.651
P53_HUMAN_Giacomelli_NULL_Etoposide_2018	5000	0	0.597	0.471	32	0.611	0.611
P53_HUMAN_Giacomelli_NULL_Nutlin_2018	5000	31	0.637	0.634	14	0.618	0.613
P53_HUMAN_Giacomelli_WT_Nutlin_2018	5000	31	0.812	0.810	28	0.830	0.826
P53_HUMAN_Kotler_2018	1048	31	0.684	0.594	30	0.693	0.675
P84126_THETH_Chan_2017	1519	32	0.497	0.497	31	0.457	0.448
PABP_YEAST_Melamed_2013	5000	10	0.673	0.637	32	0.675	0.675
PTEN_HUMAN_Matreyek_2021	5000	30	0.559	0.553	27	0.597	0.575
PTEN_HUMAN_Mighell_2018	5000	30	0.536	0.510	31	0.539	0.529
Q59976_STRSQ_Romero_2015	2999	30	0.622	0.597	32	0.665	0.665
RASH_HUMAN_Bandaru_2017	3134	19	0.494	0.448	26	0.583	0.524
RL401_YEAST_Mavor_2016	1253	13	0.520	0.448	22	0.601	0.513
RL401_YEAST_Roscoe_2013	1195	30	0.603	0.532	17	0.629	0.484
RL401_YEAST_Roscoe_2014	1380	0	0.382	0.317	24	0.532	0.448
SC6A4_HUMAN_Young_2021	5000	30	0.569	0.552	32	0.569	0.569
SPG1_STRSG_Olson_2014	5000	21	0.669	0.653	32	0.672	0.672
SRC_HUMAN_Ahler_CD_2019	3372	30	0.504	0.417	32	0.573	0.573
SUMO1_HUMAN>Weile_2017	1700	30	0.414	0.414	32	0.447	0.447
SYUA_HUMAN_Newberry_2020	2497	13	0.537	0.503	10	0.546	0.538
TADBP_HUMAN_Bolognesi_2019	1196	0	0.712	0.582	0	0.689	0.679
TPK1_HUMAN>Weile_2017	3181	32	0.251	0.251	19	0.220	0.199
TPMT_HUMAN_Matreyek_2018	3648	30	0.552	0.502	23	0.541	0.518
TPOR_HUMAN_Bridgford_S505N_2020	562	7	0.400	0.236	12	0.481	0.387
TRPC_SACS2_Chan_2017	1519	32	0.542	0.542	31	0.560	0.530
TRPC_THEMA_Chan_2017	1519	32	0.358	0.358	7	0.372	0.279
UBC9_HUMAN>Weile_2017	2563	30	0.392	0.360	31	0.451	0.448
VKOR1_HUMAN_Chiasson_abundance_2020	2695	30	0.498	0.463	30	0.511	0.462
VKOR1_HUMAN_Chiasson_activity_2020	697	32	0.418	0.418	9	0.441	0.392
YAP1_HUMAN_Araya_2012	5000	32	0.516	0.516	28	0.531	0.513

S4. Best-versus-final equivalence tests

Table 12: Best-versus-final-layer equivalence tests under the leakage-free splits (margin ± 0.01). Verdict rule: *Equivalent* when the bootstrap 90% CI of the paired mean difference lies entirely inside the margin, *Difference* when entirely outside, *Inconclusive* otherwise. The best layer is the argmax of the same per-dataset layer-mean curve (selection favours detecting a difference, so equivalence verdicts are conservative). In 7 further model-arm-split combinations the best layer is layer 32 itself (no test defined): ESM-2 masked_wt and unmasked_wt (all three splits each) and SaProt (mask) mutant (contiguous). Result: 8 equivalent, 12 inconclusive, 0 differences.

Model	Arm	Split	Best	n	Mean Δ	90% CI	TOST p	Verdict
ESM-2	mutant	Modulo	L31	63	+0.0041	[-0.0015, +0.0099]	0.049	Equivalent
ESM-2	mutant	Contiguous	L31	57	+0.0058	[-0.0040, +0.0163]	0.250	Inconclusive
ESM-2	mutant	Pos.-grouped	L31	63	+0.0029	[-0.0045, +0.0102]	0.059	Inconclusive
SaProt (mask)	masked_wt	Modulo	L30	63	+0.0054	[+0.0011, +0.0098]	0.045	Equivalent
SaProt (mask)	masked_wt	Contiguous	L31	57	+0.0077	[+0.0034, +0.0119]	0.188	Inconclusive
SaProt (mask)	masked_wt	Pos.-grouped	L31	63	+0.0035	[+0.0003, +0.0066]	0.001	Equivalent
SaProt (mask)	unmasked_wt	Modulo	L30	63	+0.0063	[+0.0010, +0.0118]	0.136	Inconclusive
SaProt (mask)	unmasked_wt	Contiguous	L31	57	+0.0041	[-0.0013, +0.0096]	0.041	Equivalent
SaProt (mask)	unmasked_wt	Pos.-grouped	L27	63	+0.0031	[-0.0038, +0.0099]	0.052	Equivalent
SaProt (mask)	mutant	Modulo	L30	63	+0.0047	[-0.0019, +0.0114]	0.098	Inconclusive
SaProt (mask)	mutant	Pos.-grouped	L30	63	+0.0038	[-0.0042, +0.0122]	0.110	Inconclusive
SaProt (3Di)	masked_wt	Modulo	L31	63	+0.0017	[-0.0010, +0.0046]	0.000	Equivalent
SaProt (3Di)	masked_wt	Contiguous	L31	57	+0.0009	[-0.0039, +0.0057]	0.001	Equivalent
SaProt (3Di)	masked_wt	Pos.-grouped	L31	63	+0.0016	[-0.0011, +0.0043]	0.000	Equivalent
SaProt (3Di)	unmasked_wt	Modulo	L31	63	+0.0071	[+0.0039, +0.0106]	0.081	Inconclusive
SaProt (3Di)	unmasked_wt	Contiguous	L31	57	+0.0086	[+0.0036, +0.0138]	0.328	Inconclusive
SaProt (3Di)	unmasked_wt	Pos.-grouped	L31	63	+0.0083	[+0.0045, +0.0126]	0.254	Inconclusive
SaProt (3Di)	mutant	Modulo	L31	63	+0.0060	[+0.0010, +0.0112]	0.101	Inconclusive
SaProt (3Di)	mutant	Contiguous	L31	57	+0.0057	[-0.0031, +0.0148]	0.218	Inconclusive
SaProt (3Di)	mutant	Pos.-grouped	L31	63	+0.0078	[+0.0029, +0.0129]	0.240	Inconclusive

S5. Foldseek 3Di token preparation disclosure

Table 13: 3Di files: length and gap-padded residue count per dataset.

Dataset	3Di length	Gaps	Dataset	3Di length	Gaps
A0A1H9GEU1_NEIME_Kennouche_2019	161	0	MSH2_HUMAN_Jia_2020	934	0
A4GRB6_PSEAI_Chen_2020	266	0	MTH3_HAEAE_Rockah-Shmuel_2015	330	0
A4_HUMAN_Seuma_2021	770	0	NUD15_HUMAN_Suiter_2020	164	0
AACC1_PSEAI_Dandage_2018	177	0	P53_HUMAN_Giacomelli_NULL_Etoposide_2018	393	0
ADRB2_HUMAN_Jones_2020	413	0	P53_HUMAN_Giacomelli_NULL_Nutlin_2018	393	0
AMIE_PSEAE_Wrenbeck_2017	346	0	P53_HUMAN_Giacomelli_WT_Nutlin_2018	393	0
B3VI55_LIPST_Klesmith_2015	439	0	P53_HUMAN_Kotler_2018	393	0
BLAT_ECOLX_Deng_2012	286	0	P84126_THETH_Chan_2017	254	0
BLAT_ECOLX_Firnberg_2014	286	0	PABP_YEAST_Melamed_2013	577	0
BLAT_ECOLX_Jacquier_2013	286	0	PTEN_HUMAN_Matreyek_2021	403	0
BLAT_ECOLX_Stiffler_2015	286	0	PTEN_HUMAN_Mighell_2018	403	0
CALM1_HUMAN_Weile_2017	149	0	Q59976_STRSQ_Romero_2015	501	22
CCDB_ECOLI_Adkar_2012	101	0	RASH_HUMAN_Bandarū_2017	189	0
CCDB_ECOLI_Tripathi_2016	101	0	RL401_YEAST_Mavor_2016	128	0
CP2C9_HUMAN_Amorosi_abundance_2021	490	0	RL401_YEAST_Roscoe_2013	128	0
CP2C9_HUMAN_Amorosi_activity_2021	490	0	RL401_YEAST_Roscoe_2014	128	0
DLG4_HUMAN_Faure_2021	724	0	SC6A4_HUMAN_Young_2021	630	0
DLG4_RAT_McLaughlin_2012	724	0	SPG1_STRSG_Olson_2014	448	0
DYR_ECOLI_Thompson_plusLon_2019	159	0	SRC_HUMAN_Ahler_CD_2019	536	0
ESTA_BACSU_Nutschel_2020	212	0	SUMO1_HUMAN_Weile_2017	101	0
F7YBW8_MESOW_Aakre_2015	93	0	SYUA_HUMAN_Newberry_2020	140	0
GAL4_YEAST_Kitzman_2015	881	0	TADBP_HUMAN_Bolognesi_2019	414	0
GCN4_YEAST_Staller_induction_2018	281	0	TPK1_HUMAN_Weile_2017	243	0
GFP_AEQVI_Sarkisyan_2016	238	0	TPMT_HUMAN_Matreyek_2018	245	0
GRB2_HUMAN_Faure_2021	217	0	TPOR_HUMAN_Bridgford_S505N_2020	635	0
HIS7_YEAST_Pokusaeva_2019	220	0	TRPC_SACS2_Chan_2017	248	0
HSP82_YEAST_Flynn_2019	709	0	TRPC_THEMA_Chan_2017	252	0
HSP82_YEAST_Mishra_2016	709	0	UBC9_HUMAN_Weile_2017	159	1
IF1_ECOLI_Kelsic_2016	72	0	VKOR1_HUMAN_Chiasson_abundance_2020	163	0
KK2_KLEPN_Melnikov_2014	264	0	VKOR1_HUMAN_Chiasson_activity_2020	163	0
MK01_HUMAN_Brenan_2016	360	0	YAP1_HUMAN_Araya_2012	504	0

S6. Fusion versus last layer: all paired comparisons

Table 14: Paired per-dataset differences (strategy – last layer; mutant arm, combined features, 20 datasets per model) for every split and readout. Cells: mean $\Delta\rho$ with bootstrap 90% CI; † marks intervals excluding zero.

Model	Split	Readout	Strategy	$\Delta\rho$	90% CI	Wilcoxon p
ESM-2	Contiguous	lightgbm	attention	-0.0508†	[-0.0759, -0.0292]	0.00107
ESM-2	Contiguous	mlp	attention	-0.0540†	[-0.0828, -0.0280]	0.00385
ESM-2	Contiguous	rf	attention	-0.0615†	[-0.0891, -0.0370]	0.00029
ESM-2	Contiguous	ridge	attention	-0.0424†	[-0.0703, -0.0160]	0.0348
ESM-2	Contiguous	lightgbm	concat_5L	-0.0320†	[-0.0449, -0.0194]	0.00258
ESM-2	Contiguous	mlp	concat_5L	-0.0990†	[-0.1288, -0.0708]	1.53e-05
ESM-2	Contiguous	rf	concat_5L	-0.0525†	[-0.0696, -0.0362]	0.000153
ESM-2	Contiguous	ridge	concat_5L	-0.0099	[-0.0552, +0.0365]	0.644
ESM-2	Contiguous	lightgbm	mean_20_33	-0.0839†	[-0.1063, -0.0630]	4.58e-05
ESM-2	Contiguous	mlp	mean_20_33	-0.0680†	[-0.0880, -0.0492]	1.53e-05
ESM-2	Contiguous	rf	mean_20_33	-0.0940†	[-0.1149, -0.0731]	4.58e-05
ESM-2	Contiguous	ridge	mean_20_33	-0.0507†	[-0.0787, -0.0245]	0.00934
ESM-2	Modulo	lightgbm	attention	-0.0453†	[-0.0862, -0.0115]	0.154
ESM-2	Modulo	mlp	attention	-0.0398†	[-0.0642, -0.0154]	0.00639
ESM-2	Modulo	rf	attention	-0.0570†	[-0.0986, -0.0225]	0.00199
ESM-2	Modulo	ridge	attention	-0.0236	[-0.0529, +0.0024]	0.43
ESM-2	Modulo	lightgbm	concat_5L	-0.0130†	[-0.0252, -0.0022]	0.0897
ESM-2	Modulo	mlp	concat_5L	-0.0447†	[-0.0686, -0.0196]	0.0107
ESM-2	Modulo	rf	concat_5L	-0.0231†	[-0.0309, -0.0164]	1.91e-06
ESM-2	Modulo	ridge	concat_5L	+0.0177	[-0.0079, +0.0437]	0.33
ESM-2	Modulo	lightgbm	mean_20_33	-0.0503†	[-0.0650, -0.0350]	8.2e-05
ESM-2	Modulo	mlp	mean_20_33	-0.0439†	[-0.0645, -0.0239]	0.00271
ESM-2	Modulo	rf	mean_20_33	-0.0560†	[-0.0700, -0.0424]	1.91e-06
ESM-2	Modulo	ridge	mean_20_33	+0.0101	[-0.0011, +0.0229]	0.349
ESM-2	Random	lightgbm	attention	-0.0023	[-0.0100, +0.0048]	0.812
ESM-2	Random	mlp	attention	+0.0069	[-0.0080, +0.0234]	0.701
ESM-2	Random	rf	attention	-0.0055	[-0.0184, +0.0053]	0.452
ESM-2	Random	ridge	attention	-0.0019	[-0.0107, +0.0073]	0.812
ESM-2	Random	lightgbm	concat_5L	+0.0088†	[+0.0055, +0.0122]	0.000322
ESM-2	Random	mlp	concat_5L	-0.0192†	[-0.0296, -0.0082]	0.00558
ESM-2	Random	rf	concat_5L	+0.0091†	[+0.0054, +0.0128]	0.00169
ESM-2	Random	ridge	concat_5L	+0.0124†	[+0.0052, +0.0210]	0.000483
ESM-2	Random	lightgbm	mean_20_33	-0.0015	[-0.0050, +0.0017]	0.729
ESM-2	Random	mlp	mean_20_33	-0.0040	[-0.0102, +0.0021]	0.33
ESM-2	Random	rf	mean_20_33	-0.0017	[-0.0074, +0.0044]	0.368
ESM-2	Random	ridge	mean_20_33	-0.0009	[-0.0038, +0.0017]	0.956
SaProt	Contiguous	lightgbm	attention	-0.0276†	[-0.0427, -0.0115]	0.0232
SaProt	Contiguous	mlp	attention	-0.0228†	[-0.0384, -0.0066]	0.0232
SaProt	Contiguous	rf	attention	-0.0254†	[-0.0448, -0.0003]	0.00665
SaProt	Contiguous	ridge	attention	-0.0185	[-0.0386, +0.0008]	0.174
SaProt	Contiguous	lightgbm	concat_5L	-0.0293†	[-0.0417, -0.0160]	0.00316
SaProt	Contiguous	mlp	concat_5L	-0.0495†	[-0.0656, -0.0335]	0.00029
SaProt	Contiguous	rf	concat_5L	-0.0352†	[-0.0494, -0.0193]	0.00316
SaProt	Contiguous	ridge	concat_5L	-0.0366†	[-0.0569, -0.0159]	0.0174
SaProt	Contiguous	lightgbm	mean_20_33	-0.0297†	[-0.0425, -0.0178]	0.00107
SaProt	Contiguous	mlp	mean_20_33	-0.0109	[-0.0264, +0.0058]	0.0395
SaProt	Contiguous	rf	mean_20_33	-0.0243†	[-0.0361, -0.0116]	0.011
SaProt	Contiguous	ridge	mean_20_33	-0.0100	[-0.0254, +0.0055]	0.284
SaProt	Modulo	lightgbm	attention	-0.0145†	[-0.0275, -0.0019]	0.105
SaProt	Modulo	mlp	attention	-0.0193†	[-0.0295, -0.0091]	0.0073
SaProt	Modulo	rf	attention	-0.0094	[-0.0195, +0.0007]	0.105
SaProt	Modulo	ridge	attention	-0.0023	[-0.0124, +0.0074]	0.756
SaProt	Modulo	lightgbm	concat_5L	-0.0120†	[-0.0177, -0.0064]	0.00365
SaProt	Modulo	mlp	concat_5L	-0.0388†	[-0.0565, -0.0219]	0.00233
SaProt	Modulo	rf	concat_5L	-0.0082	[-0.0160, +0.0000]	0.0696
SaProt	Modulo	ridge	concat_5L	+0.0028	[-0.0160, +0.0207]	0.674
SaProt	Modulo	lightgbm	mean_20_33	-0.0087	[-0.0182, +0.0006]	0.165
SaProt	Modulo	mlp	mean_20_33	0.0164†	[0.0293, 0.0035]	0.0532

S7. Readout contrast on the last layer (LightGBM vs Ridge)

Table 15: Paired per-dataset differences (LightGBM – Ridge) on the last layer (mutant arm, combined features, 20 datasets per model).

Model	Split	n	Mean Δ	90% CI	Wilcoxon p
ESM-2	Contiguous	17	+0.0582	[+0.0263, +0.0903]	0.0129
ESM-2	Modulo	20	+0.0449	[+0.0214, +0.0675]	0.0107
ESM-2	Random	20	-0.0351	[-0.0436, -0.0270]	1.34e-05
SaProt	Contiguous	17	+0.0535	[+0.0062, +0.1020]	0.12
SaProt	Modulo	20	+0.0160	[-0.0161, +0.0483]	0.571
SaProt	Random	20	-0.0651	[-0.0804, -0.0503]	9.54e-06

S8. Pooling means under all splits

Table 16: Pooling means (mutant arm, combined features; mean Spearman ρ over 20 datasets per model) for all splits, strategies, and readouts.

Model	Split	Strategy	Ridge	RF	MLP	LightGBM
ESM-2	Random	last_layer	0.708	0.644	0.669	0.673
ESM-2	Random	mean_20_33	0.707	0.642	0.665	0.671
ESM-2	Random	concat_5L	0.721	0.653	0.650	0.682
ESM-2	Random	attention	0.706	0.638	0.676	0.671
ESM-2	Modulo	last_layer	0.463	0.476	0.480	0.508
ESM-2	Modulo	mean_20_33	0.473	0.420	0.436	0.457
ESM-2	Modulo	concat_5L	0.481	0.453	0.435	0.495
ESM-2	Modulo	attention	0.439	0.419	0.440	0.463
ESM-2	Contiguous	last_layer	0.375	0.397	0.429	0.433
ESM-2	Contiguous	mean_20_33	0.324	0.303	0.361	0.349
ESM-2	Contiguous	concat_5L	0.365	0.345	0.330	0.401
ESM-2	Contiguous	attention	0.333	0.336	0.375	0.382
ESM-2	Pos.-grouped	last_layer	0.463	0.470	0.479	0.504
ESM-2	Pos.-grouped	mean_20_33	0.467	0.416	0.433	0.451
ESM-2	Pos.-grouped	concat_5L	0.471	0.454	0.432	0.489
ESM-2	Pos.-grouped	attention	0.428	0.418	0.442	0.456
SaProt	Random	last_layer	0.667	0.549	0.638	0.602
SaProt	Random	mean_20_33	0.670	0.587	0.655	0.632
SaProt	Random	concat_5L	0.697	0.613	0.640	0.651
SaProt	Random	attention	0.677	0.608	0.664	0.647
SaProt	Modulo	last_layer	0.432	0.415	0.445	0.448
SaProt	Modulo	mean_20_33	0.433	0.415	0.429	0.439
SaProt	Modulo	concat_5L	0.434	0.407	0.406	0.436
SaProt	Modulo	attention	0.429	0.405	0.426	0.433
SaProt	Contiguous	last_layer	0.287	0.308	0.318	0.341
SaProt	Contiguous	mean_20_33	0.277	0.284	0.308	0.311
SaProt	Contiguous	concat_5L	0.251	0.273	0.269	0.312
SaProt	Contiguous	attention	0.269	0.283	0.296	0.313
SaProt	Pos.-grouped	last_layer	0.427	0.413	0.431	0.444
SaProt	Pos.-grouped	mean_20_33	0.427	0.402	0.413	0.433
SaProt	Pos.-grouped	concat_5L	0.433	0.398	0.398	0.424
SaProt	Pos.-grouped	attention	0.422	0.395	0.420	0.430

S9. Layer-similarity analysis: 50 vs 200 positions, debiased CKA

Table 17: Layer-similarity summaries at 50 and 200 sampled positions (10 datasets each). The debiased (unbiased-HSIC) estimator is unstable at $n < d$ (estimates can exceed 1), so the 200-position standard estimator is the primary estimate.

Model	Adjacent		L0 vs L32		Pearson r vs distance	
	cos	CKA	cos	CKA	cos	CKA
ESM-2 (50 pos.)	0.962	0.967	+0.558	0.554	-0.93	-0.60
ESM-2 (200 pos.)	0.962	0.953	+0.542	0.409	-0.93	-0.66
debiased (200)	–	1.002	–	0.963	–	–
SaProt (50 pos.)	0.979	0.979	-0.009	0.808	-0.50	-0.68
SaProt (200 pos.)	0.979	0.969	-0.008	0.779	-0.50	-0.70
debiased (200)	–	1.001	–	1.112	–	–

S10. Published ProteinGym anchors

Table 18: Published ProteinGym v1.3 results recomputed on the same assays as this work (46 of 63 covered by the release). Zero-shot column: mean per-assay Spearman ρ over the 46 assays; our lens and reproduction rows are matched to the same assays. Supervised columns: official out-of-fold predictions under the release fold schemes (random-fold values: Kermut 0.711, ProteinNPT 0.697, Emb.-aug. MSA Transformer 0.641, OHE 0.570; our Envision-style baseline scores 0.634 under random).

Zero-shot method	ρ	Supervised method	Modulo	Contiguous
ESM-2 650M (published)	0.472	Kermut	+0.605	+0.561
GEMME	0.474	ProteinNPT	+0.566	+0.525
EVE (single)	0.478	Emb.-aug. MSA Transformer	+0.572	+0.536
DeepSequence (single)	0.457	OHE, not augmented	-0.003	+0.027
MSA Transformer (single)	0.447			
ESM-1v (single)	0.445			
ESM-2 3B	0.448			
ESM-2 150M	0.421			
Our lens: ESM-2 final	0.438	Our leakage-free probing	0.32–0.53	
Our lens: SaProt final	0.440			
Our SaProt reproduction	0.484			
Envision-style supervised (ours)	–	Envision-style (Section 3.7)	0.410	0.302

S11. Single-substitution sensitivity analysis

Table 19: Single-substitution-only sensitivity analysis (mutant arm, combined features). Each variant contributes one row, so no variant-level label sharing across folds remains. Best layer per model–split with the mean paired best-minus-final-layer difference and its bootstrap 90% CI. The best layer is L30–L31 in every leakage-free column. Panel B: real-3Di versus structure-masked SaProt at the final layer on the same subset.

Model	Split	Best	Best ρ	L32 ρ	Δ	90% CI
ESM-2	Modulo	L30	0.443	0.427	+0.0159	[+0.0057, +0.0269]
ESM-2	Contiguous	L31	0.353	0.346	+0.0071	[-0.0035, +0.0180]
ESM-2	Pos.-grouped	L30	0.444	0.425	+0.0187	[+0.0036, +0.0357]
SaProt (mask)	Modulo	L30	0.433	0.428	+0.0050	[-0.0027, +0.0126]
SaProt (mask)	Contiguous	L30	0.310	0.307	+0.0031	[-0.0069, +0.0135]
SaProt (mask)	Pos.-grouped	L30	0.428	0.423	+0.0057	[-0.0044, +0.0161]
SaProt (3Di)	Modulo	L30	0.492	0.486	+0.0060	[-0.0013, +0.0133]
SaProt (3Di)	Contiguous	L31	0.449	0.439	+0.0100	[+0.0019, +0.0187]
SaProt (3Di)	Pos.-grouped	L31	0.490	0.481	+0.0088	[+0.0031, +0.0146]
<i>B. Real-3Di vs structure-masked at L32 (paired, single-site subset)</i>						
Split	n	Mean Δ	Wilcoxon p			
Random	63	+0.0280	4.30e-05			
Modulo	63	+0.0573	1.02e-03			
Contiguous	57	+0.1316	4.83e-07			
Pos.-grouped	63	+0.0579	1.10e-03			